

# TWO WEIGHT-TWENTY AFFINE ORBITS OF MINIMAL VANISHING SUMS OF DISTINCT 105TH ROOTS

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ABSTRACT. Let  $\zeta_{105} = e^{2\pi i/105}$ . We give a certified finite classification of the inclusion-minimal vanishing subsets of distinct 105th roots through weight 20, modulo translation and multiplication by a unit modulo 105. Exactly two affine-Galois orbits occur at weight 20; their canonical representatives are displayed explicitly. Completeness is reduced to exact enumeration of 1,209,813 normalized seven-tuples of equal-valued subsets of the 15th roots. A standard-library verifier reconstructs the cyclotomic arithmetic, checks every representative and orbit, and finds no unblocked tuple. The result is restricted to distinct roots and weights at most 20; no assertion is made at weight 21 or above.

## 1. INTRODUCTION

A vanishing sum of roots of unity is *minimal* when none of its nonempty proper subsums vanishes. Lam and Leung developed general group-ring methods, weight restrictions, and smallest asymmetric examples for such sums [2]. Sivek characterized the cardinalities for which a vanishing subset of distinct  $n$ -th roots exists [3]; that existence question does not impose inclusion-minimality and does not classify affine orbits. Christie, Dykema, and Klep classified general types through weight 16 and gave a computer-assisted, explicitly conjectural extension through weight 21 in the current revision [1].

For the fixed conductor 105, a dated TheoremDB computation recorded seven affine-Galois orbits through weight 19 and left weight 20 open [4]. We reproduce that boundary and settle the next weight. The novelty statement here is deliberately bounded: searches of the cited paper, its indexed citations, public code, mathematical databases, and the two exact representatives found no prior fixed-conductor answer as of August 30, 2026.

The proof is computational but exact. Its main reduction is short: after writing  $\zeta_{105} = \zeta_7 \zeta_{15}^{-2}$ , a subset vanishes precisely when seven associated subset sums of 15th roots have the same value. This converts the decisive search into a complete enumeration over  $2^{15}$  masks, with no floating-point comparison or reliance on the discovery solver's final status.

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## 2. STATEMENT AND AFFINE ACTION

Let  $G = \mathbb{Z}/105\mathbb{Z}$ , let  $G^\times$  be its unit group, and fix  $\zeta_{105} = e^{2\pi i/105}$ . A nonempty subset  $S \subseteq G$  is *vanishing* if

$$\sigma(S) := \sum_{s \in S} \zeta_{105}^s = 0,$$

and is *inclusion-minimal vanishing* if  $\sigma(T) \neq 0$  for every nonempty proper subset  $T \subsetneq S$ . The affine-Galois group acts by

$$S \longmapsto a + uS \quad (a \in G, u \in G^\times).$$

Translation multiplies  $\sigma(S)$  by  $\zeta_{105}^a$ , while multiplication by  $u$  applies a Galois automorphism. Thus vanishing and minimality are invariant. We call the lexicographically least sorted exponent tuple in an orbit its canonical representative.

**Theorem 2.1.** *Up to the affine-Galois action, there are exactly two inclusion-minimal vanishing subsets of distinct 105th roots of weight 20. Their canonical representatives are*

$$\begin{aligned} S_{20,a} &= \{0, 1, 3, 9, 11, 24, 26, 30, 41, 42, 45, 46, 61, 63, 71, 72, \\ &\quad 76, 84, 87, 93\}, \\ S_{20,b} &= \{0, 1, 3, 11, 12, 16, 24, 33, 41, 42, 45, 46, 54, 63, 71, 75, \\ &\quad 76, 84, 86, 87\}. \end{aligned}$$

*Their affine orbit sizes are respectively 210 and 315, and their stabilizer sizes in the affine group of order 5,040 are respectively 24 and 16.*

For reference, the complete list through weight 20 consists of the following nine representatives. The two labels at the same weight distinguish different orbits.

$$\begin{aligned} S_3 &= \{0, 35, 70\}, \\ S_5 &= \{0, 21, 42, 63, 84\}, \\ S_7 &= \{0, 15, 30, 45, 60, 75, 90\}, \\ S_{14} &= \{0, 1, 8, 22, 29, 30, 43, 45, 60, 64, 71, 75, 90, 92\}, \\ S_{16} &= \{0, 1, 5, 20, 22, 30, 35, 43, 45, 60, 64, 65, 75, 80, 90, 95\}, \\ S_{18,a} &= \{0, 1, 3, 11, 16, 24, 26, 41, 42, 45, 46, 61, 63, 71, 76, 84, 86, 87\}, \\ S_{18,b} &= \{0, 1, 3, 18, 22, 29, 30, 33, 45, 48, 60, 63, 64, 71, 75, 90, 92, 93\}. \end{aligned}$$

The associated orbit and stabilizer data are as follows.

| representative | orbit size | stabilizer size |
|----------------|------------|-----------------|
| $S_3$          | 35         | 144             |
| $S_5$          | 21         | 240             |
| $S_7$          | 15         | 336             |
| $S_{14}$       | 105        | 48              |
| $S_{16}$       | 105        | 48              |
| $S_{18,a}$     | 315        | 16              |
| $S_{18,b}$     | 210        | 24              |
| $S_{20,a}$     | 210        | 24              |
| $S_{20,b}$     | 315        | 16              |

### 3. THE SEVEN-FIBER REDUCTION

The usual cyclotomic criterion already makes vanishing exact. If  $P_S(X) = \sum_{s \in S} X^s$ , then

$$(1) \quad P_S(\zeta_{105}) = 0 \iff \Phi_{105}(X) \mid P_S(X) \iff P_S(X) \bmod \Phi_{105}(X) = 0.$$

The residue has 48 integer coordinates. The following finer criterion is what makes a compact exhaustive certificate possible.

**Lemma 3.1** (Seven-fiber lemma). *Put  $\alpha = e^{2\pi i/7}$  and  $\beta = e^{2\pi i/15}$ . For each  $i \in \mathbb{Z}/7\mathbb{Z}$ , define*

$$M_i(S) = \{-2s \bmod 15 : s \in S, s \equiv i \pmod{7}\}, \quad F_i(S) = \sum_{j \in M_i(S)} \beta^j.$$

Then

$$\sigma(S) = 0 \iff F_0(S) = F_1(S) = \cdots = F_6(S).$$

Moreover, the seven masks  $M_i(S)$  determine  $S$  uniquely.

*Proof.* The identity  $1/7 - 2/15 = 1/105$  gives  $\zeta_{105} = \alpha\beta^{-2}$ , whence

$$\sigma(S) = \sum_{i=0}^6 \alpha^i F_i(S).$$

Let  $K = \mathbb{Q}(\beta)$ . Since  $\mathbb{Q}(\alpha, \beta) = \mathbb{Q}(\zeta_{105})$ , we have  $[\mathbb{Q}(\alpha, \beta) : K] = 48/8 = 6$ . Thus the minimal polynomial of  $\alpha$  over  $K$  is  $\Phi_7(T) = 1 + T + \cdots + T^6$ . Subtracting  $F_6(S)\Phi_7(\alpha) = 0$  shows that  $\sigma(S) = 0$  is equivalent to

$$\sum_{i=0}^5 (F_i(S) - F_6(S))\alpha^i = 0.$$

The elements  $1, \alpha, \dots, \alpha^5$  are a  $K$ -basis, so all seven values are equal. Finally, the map

$$s \longmapsto (s \bmod 7, -2s \bmod 15)$$

is a bijection by the Chinese remainder theorem, proving uniqueness.  $\square$

## 4. EXACT FINITE CERTIFICATE

For a mask  $M \subseteq \mathbb{Z}/15\mathbb{Z}$ , reduce its mask polynomial exactly:

$$R_M(X) = \left( \sum_{j \in M} X^j \right) \bmod \Phi_{15}(X).$$

Represent  $M$  by the eight integer coefficients of  $R_M$ , where

$$\Phi_{15}(X) = X^8 - X^7 + X^5 - X^4 + X^3 - X + 1.$$

Two masks have the same value at  $\beta$  exactly when these coefficient vectors agree. The following proposition is the decisive finite statement.

**Proposition 4.1** (Exhaustive computation). *The 32,768 masks of  $\mathbb{Z}/15\mathbb{Z}$  form 14,221 exact value classes. Among all ordered seven-tuples of masks from a common value class, exactly 1,209,813 have total cardinality at most 20 and contain the coordinate pair corresponding to exponent 0. Every one of those tuples contains an affine image of one of the nine representatives displayed above. There are 1,331 distinct such affine images, and the number of unblocked tuples is zero.*

*Certificate and verification.* The builder groups all masks by their integer residue vectors, then visits value classes lexicographically and masks numerically. In the first fiber it requires bit 0, which is precisely the normalization  $0 \in S$ . A recursive integer weight bound visits every and only the seven-tuples in the statement. For each reconstructed subset it records the least contained blocker in the fixed ordering by weight and exponent tuple.

The serialized manifest records the following count by total weight:

| weight | tuples | weight | tuples  |
|--------|--------|--------|---------|
| 3      | 1      | 12     | 6,008   |
| 5      | 1      | 13     | 1,365   |
| 6      | 34     | 14     | 11,396  |
| 7      | 1      | 15     | 47,285  |
| 8      | 48     | 16     | 17,536  |
| 9      | 561    | 17     | 94,180  |
| 10     | 60     | 18     | 293,142 |
| 11     | 957    | 19     | 150,518 |
|        |        | 20     | 586,720 |

The tuple-stream SHA-256 is

620fbdb4bf9e1a8ac402b3b81208ef26c72915e0a3dc68af2fc85bc4f908ccca,

and the first-blocker witness-stream SHA-256 is

11f1684b3dcfe4d20edcfca40e37a3c6d3e5ab07ee73a043a31891335497ea50.

The release verifier imports neither the builder nor the Z3 discovery program. Using only the Python standard library, it reconstructs  $\Phi_{105}$  by exact monic division, rebuilds all monomial residues, checks the matrix digest, and verifies every listed sum. It checks inclusion-minimality independently by meet-in-the-middle integer subset sums. It also expands all 5,040 affine maps, checks canonicity, orbit sizes, stabilizers, and disjointness, and rebuilds the 1,331 blockers. For the fiber computation it obtains all mask values by a Gray-code walk, rather than the builder’s subset dynamic program, and replays the complete enumeration. The reconstructed manifest agrees field-for-field with the recorded invariants and has SHA-256

886b29e4c3f149b66f9ce6f9a0ad1ada63a31f08194af4495fd400d51f63519c.

In particular, it independently recomputes the unblocked count as zero.  $\square$

*Proof of Theorem 2.1.* First, the exact cyclotomic and meet-in-the-middle checks in Proposition 4.1 prove that every displayed representative is vanishing and inclusion-minimal.

Conversely, let  $S$  be a nonempty inclusion-minimal vanishing subset with  $|S| \leq 20$ . Translate  $S$  so that  $0 \in S$ . Lemma 3.1 places the resulting set among the 1,209,813 tuples exhaustively checked in Proposition 4.1. Hence it contains a listed affine image  $B$ . The exact checks show that  $B$  is itself a nonempty vanishing subset. Minimality of  $S$  forces  $S = B$ . Thus the nine displayed orbits are the complete list through weight 20, and selecting the two weight-20 entries proves the theorem.  $\square$

## 5. REPRODUCIBILITY, AUDIT, AND LIMITATIONS

From the project root, the fast fail-closed tests and the full independent verification are respectively

```
python3 tests/test_verifier_fail_closed.py
python3 certificates/verify_weight20.py \
  certificates/weight20_certificate.json
```

The full run recomputes 1,209,813 tuples and prints ‘status=VERIFIED’ with zero unblocked tuples. The cyclotomic matrix SHA-256 is

669ac026c32d1396ec85f4a7348a848918221d9a9c19598428011eefcfcc2744.

All decisive arithmetic is over integers. The discovery stage used SymPy 1.14.0 and Z3 4.15.3.0, but the certificate verifier requires neither package. Mutation tests reject malformed schemas, wrong conductors, truncated digests, path traversal, changed representatives, and corrupted manifest bindings.

The theorem concerns subsets, hence distinct roots and coefficients in  $\{0, 1\}$ ; it does not classify multisets or general integer coefficients. The empty set is excluded, and weight 20 is included. No claim is made for weights at least 21 or for the complete unbounded conductor-105 problem. No proof assistant was used.

## COMPUTATIONAL AND AI DISCLOSURE

OpenAI Codex was used for literature searching, code generation and execution, exact-certificate construction, adversarial testing, and manuscript drafting. The named author is solely responsible for the mathematical claims and the released artifacts. No floating-point computation contributes to the proof.

## REFERENCES

1. Louis Christie, Kenneth J. Dykema, and Igor Klep, *Classifying minimal vanishing sums of roots of unity*, 2025, arXiv:2008.11268v2, <https://arxiv.org/abs/2008.11268>.
2. T.Y Lam and K.H Leung, *On vanishing sums of roots of unity*, *Journal of Algebra* **224** (2000), no. 1, 91–109.
3. Gary Sivek, *On vanishing sums of distinct roots of unity*, *Integers* **10** (2010), no. 3, 365–368.
4. TheoremDB contributors, *Minimal vanishing sums of distinct 105th roots*, TheoremDB research memory, snapshot of July 28, 2026, 2026, Available at <https://theoremdb.org/statements/minimal-vanishing-105th-root-sums>; accessed August 30, 2026.

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